INFINITY RACING
ESTD. 2022

aramco
F1 in Schools
SINGAPORE | 2023
WORLD FINALS

DESIGN & ENGINEERING PORTFOLIO



LUCKY
ENCOURAGING INNOVATION

RAYO RACING
INFINITY Premium Motoring CARS

Ansys
HOP HOUSE OF POLISH

KeyShot
FORGE 3D

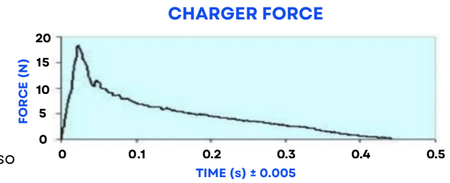
ABEC 3 5 7

INITIAL RESEARCH & CONCEPTS

1.1 /// THRUST EFFICIENCY

It was essential for us to understand when the car would be getting the maximum amount of thrust force from the CO2 chamber canister. At this point the efficiency of the gas canister would also be the greatest. From the above research we were able to gain 2 key findings :

- It was critical for us to find out the point at which the energy was maximum so we could particularly look into that moment and maximise the car's performance at that time.
- The limited energy obtained from the canister must be preserved for as long as possible, and any loss of it must be minimised.



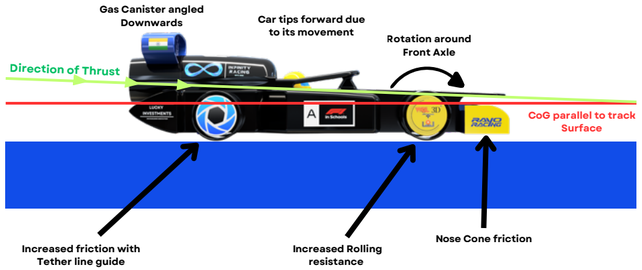
/// *Fig 1.1 | Force of canister over time graph (Credit: EESW STEM Cymru)*

TIPPING MOMENT

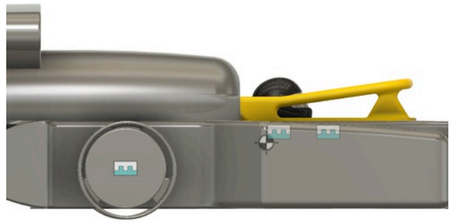
Firstly to implement modifications in the car for the above pointers, it was important for us to find a way in which we were able to notice this happening. To do this, our team recorded a slow motion video of the car in action on the race track. We observed that the car had its maximum acceleration during the launch phase period. As the car goes down the track the velocity decreases hence the deceleration increases.

From the formula derived in Newton's second law of motion, F=ma, we could derive when the acceleration of the car would be the greatest so would the thrust force. Hence the thrust force was the greatest at the launch phase period. Our team specifically observed a recording of the car at the launch phase and we saw that as the car was launched the rear wheels of the car would lift off the ground and multiple components of the car would vibrate due to the force of the canister. This took place because of an offset between the centre of gravity and the large thrust force.

This would be undesirable, because when the car tipped forward the car was producing a forward thrust at a downward angle. This was not only causing some of the energy to be used up inefficiently but it also increased rolling resistance of the front wheels and the friction in the tether line guides. Hence, it was very important for the team to reduce the tipping moment of the car.



/// *Fig 1.2 | Diagram showing events occurring during launch period (Credit: Infinity Racing)*



/// *Fig 1.3 | Image showing position of CoG to maximise phase effect (Credit: Infinity Racing)*

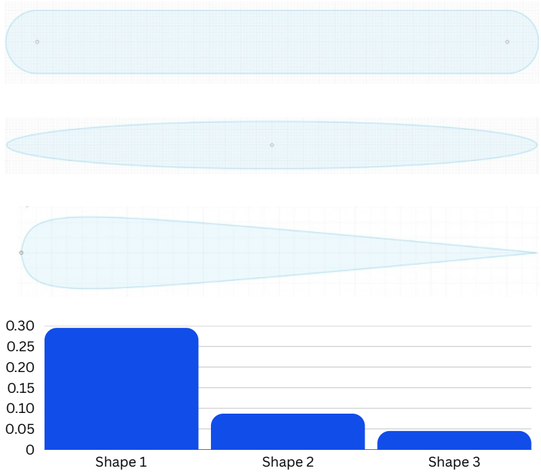
APPLICATION

To reduce the tipping moment, we had to use the formula, Torque=Force*Perpendicular distance. So from the formula it can be seen that by decreasing the perpendicular distance of the CoG, the moment will be reduced. To do this we had to align the CoG and the thrust vector closely. Along with this to minimise the effect of the tipping moment, the weight of the restoring moment can be increased by placing the CoG more rearwards. Therefore, our goal is to align the CoG and thrust vector as closely as possible and make sure that the CoG is at the rear end but also make sure that to not change it so much that it would hamper with the balance of the car during the other phases of the race.

1.2 /// WING DESIGN

The wings of the car took up a combined 15% of the mass of the car. It was very important for us to pay close attention to the shape of the wings since they took up such a huge mass of the car and would greatly impact its aerodynamics.

To check the effect of different shapes on the aerodynamics of the car, our team decided to take 3 shapes which would be beneficial to the car aerodynamically and ran air flow simulations on those shapes to check out how each of them would direct the wind.



/// *Fig 1.4 | Graph showing drag coefficients of different components (Credit: Infinity Racing)*

WING TYPE: AIRFOIL

The airfoil shape was one that we had incorporated in the national finals and for the World Finals we decided to delve deeper into the exact benefits of it. We ran numerous simulations using this design and also did a great amount of research about this concept. Using all of this data, we managed to come up with a concrete conclusion about the airfoil shape.

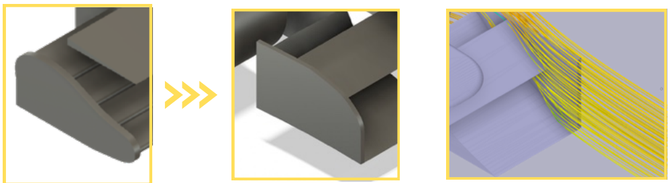
The main purpose of a front wing is to generate downforce, and the airfoil shape greatly aids in doing so. This shape helps to create a region of high pressure below the wing and low pressure above it, thus resulting in a net downward force. Downforce helps to keep the car firmly planted on the track.

The airfoil shape also helps to reduce lift, which is the upward force which can cause the car to almost become airborne if not moderated. By creating a high pressure zone above the wing and low pressure below it, the airfoil shape counters the lifting force acting on the car and ensures optimal traction and grip.

Airfoil shape on the front wing also helps to optimise the car's aerodynamic performance. It helps in reducing drag by smoothly guiding the airflow around the car, minimising the amount of air pockets created that may create resistance. This allows the car to cut through the air more efficiently, resulting in higher speeds.

1.3 /// ENDPLATES

Our team was inspired by the actual F1 cars in which endplates are present on the sides of the wings of the car. In Formula 1 cars endplates are used to redirect the airflow hitting the endplate around the tyres. This minimises the drag resistance faced by the car and directs the air to flow to the side pods and to the car floor. Our team had first seen the advantages of using endplates at the wings of our car in the 2022 Nationals, however we were not able to incorporate an effective design of the end plates into the car. The endplates at the rear wing were seen to be a highly effective and innovative design which supported many of the scientific theories regarding aerodynamics too. The benefits of this design were seen to be highly impactful as it would reduce the drag produced greatly but at the same time it would also indulge in an increase of weight and frontal surface area of the car. Hence it was very important to get the best design of the endplates for our car and we chose to go for the angled deflector which had been used in many actual F1 cars.



/// *Fig 1.5 | Evaluation and development of endplates (Credit: Infinity Racing)*

1.4 /// CATAMARAN CONCEPT

A catamaran is a boat which has twin hulls parallel to each other. We believed this boat design could also be incorporated in our car's floor design as it had multiple benefits.

- It enhanced our cars aerodynamics as it reduced drag and also significantly improved the airflow under the car.
- The dual hull structure helps the car massively in terms of stability. The weight of the car is evenly distributed by this structure, hence making it far more balanced.
- This design helped to generate just the right amount of downforce our car required.

The incorporation of a catamaran concept, enhanced the pace of our car on tracks by margins which we did not expect from such a simple concept.

EVALUATION

From the various concepts we learned from our initial research we were able to summarise our car's design objectives which were :

1. Keep the mass of the car as low as possible
2. The car's dimensions should comply with the regulations
3. To reduce the initial tipping moment at the start of the race
4. The car should be durable enough to not be damaged during races

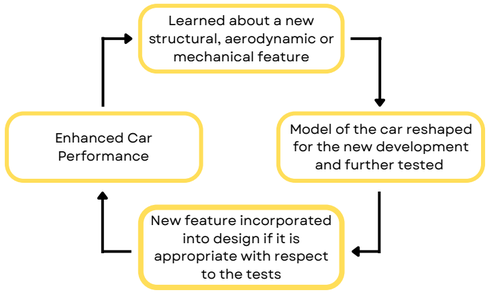
Based on the CFD results, we found that our design concepts for components such as the endplate and wing shape significantly improved the car's aerodynamics.

Improvement Actions : Increasing mass of car at rear end, incorporating NACA 0015 shape in wings, increase in angle of endplates to deflect air to the side.

2.1 /// FORMULA OF DEVELOPMENT

Research and Development, according to our team, is the most important stage in the design process of our car. It was essential for us to appropriately carry out the actions involved in this stage. To enhance the development process of the car with our research, we had devised our own formula, 'The Formula of development' to aid us in perfecting this stage. The formula of development had been devised for the India National Finals round and proved to be extremely helpful to the team. Hence, the team decided to implement the formula for the World Finals too. This formula primarily dictated the steps to be taken when making any aerodynamic, mechanical or structural modification to the car. It was important for us to understand that if we changed one part of the car it would affect the whole car's performance, sometimes in an adverse way.

Step 1: The team's design engineers research on the various scientific theories & principles that affect the car's performance and study the results of simulations run on the car. If they discover something which could improve the car's performance, it is discussed with the team.



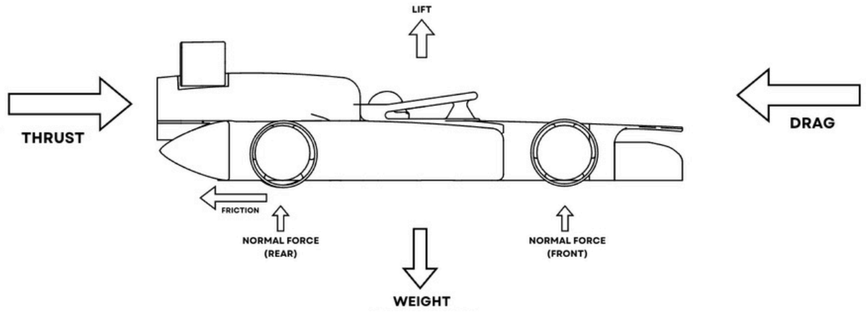
Step 2: The CAD model of the feature is made or the car is tweaked to incorporate it. Then the new feature is tested independently as well as with the whole car to check whether it is enhancing the car's performance.

Step 3 : If the tests conducted on the new feature of the car show positive results then it is incorporated into the car design.

Step 4: After appropriately carrying out all these steps, the speed of the car is improved. To ensure this the whole car is retested to see if the drag coefficient has decreased or the performance has improved according to the simulations.

2.2 /// FORCES IN PLAY

From our research, we saw that F1 in schools was mainly a Drag race in which the forces affecting the car had the most significant impact on the car's speed. Hence it was important for us to understand the effect of each of these forces and make the necessary modifications the car design. To make a note of all the forces we created a free body diagram of the forces in play.



/// Fig 2.1 | Free body diagram showing effect of forces on car (Credit: Infinity Racing

/// DRAG:

Drag is the force which resists the motion of a moving body in a medium. To minimise the drag force, we had to pay attention to each of the forms of the drag force. The main forms of drag affecting our car are : form drag and skin friction drag.

- *Skin Friction Drag:* This type of drag is influenced by the surface texture of the car's body and components. Smoother surfaces of the car would experience less skin friction drag compared to the rougher parts. Even the smoothness of the material could have an impact on this type of drag.
- *Form Drag:* Form drag, also referred to as pressure drag, is influenced by the shape and size of the car and its components. A streamlined and aerodynamically efficient car shape can help reduce the pressure drag and improve the car's performance.

/// UNDERSTANDING THE DRAG EQUATION

$$F_d = \frac{1}{2}\rho u^2 C_d A$$

SYMBOL	DESCRIPTION	UNDER CONTROL?
ρ	Density of surrounding air	No
v	Velocity of car	No
Cd	Drag coefficient	Yes
A	Frontal Area (In contact with air)	Yes

/// IMPROVISATION ON CAR

- To ensure that the skin friction drag was the least possible, we ensured to sand the car appropriately. Then add 2 layers of primer, then the paint and lacquer to ensure a smooth layer on the car's surface.
- The front and rear wing were manufactured by a SLA printing method to ensure a smooth finish of the front and rear wing reducing skin friction drag.
- The frontal area of the front wing was kept minimum to reduce the drag force. The body was kept streamlined so that the area in contact with the frontal, incoming air was reduced. The width of the front wheels were kept minimum at 13-15mm ,according to the regulations.

/// LIFT & DOWNFORCE

Understanding Lift:
Lift refers to the upward force faced by the car as it moves through the air. It is critical to minimise the lift force of the car. An increased lift force causes the wheels of the car to lift off the ground, which results in reduced traction and stability of the car. This will reduce the acceleration of the car and affect the overall car's performance.

$$F = Cl \times q \times A$$

From this equation we can see that, two of the three variables, the lift coefficient and area of the front wing were in our control.

Understanding Downforce:
Downforce is the force that acts on the car in the downwards direction, parallel to the direction of the airflow. Downforce helps in increasing the traction of the wheel with the track surface which increases the stability of the car. But it was important for us to make sure that we do not create excess downforce which would increase the friction of the car and could also lead to the deformation of the wheels. From our testing, we saw that changing the angle of the front wing would affect the downforce generation.

/// ANGLE OF ATTACK

From our virtual testing, we saw that increasing the angle of attack of the front will increase the downforce generated by the car and would increase the drag. Hence we increased the angle of attack by 2 degrees to increase the drag of the car. To find the optimal angle where the drag and downforce was balanced.

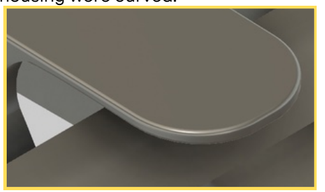
	DESIGN 1	DESIGN 2
ANGLE (DEGREES)	1	3
DRAG (N)	0.039	0.045

2.3 /// COANDA EFFECT

It refers to the tendency of fluids to remain attached to curved surfaces. This indicates that there will be smooth airflow around curved surfaces, which leads to laminar airflow around the car which helps in reducing the drag force faced by the car, in turn improving the straight-line speed of the car. Whereas angular surfaces tend to create turbulent airflows.

Conclusion:
We had to maximise the number of curved surfaces on the car to avoid a turbulent airflow around the car, which would have increased the drag.

Improvisation:
- A minimum fillet of radius 0.25mm was applied to all the edges of the front wing and body. Rounding these edges ensured that the air would stick to them.
- Maximum surface area of the car was covered by the part of the body covering the CO2 chamber. To increase the effect of coanda effect in this region, it was ensured that all the surfaces around the CO2 chamber housing were curved.




/// Fig 2.2 | Evidence of use of Fillets (Credit: Infinity Racing)

2.4 /// BERNOULLI'S PRINCIPLE

It is a fundamental concept in fluid dynamics and demonstrates the relation between pressure and velocity.

Understanding the concept :

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = constant$$

Where
- p is density
- P is pressure
- v is velocity

Hence. from this we can see that velocity and pressure are inversely proportional. So, the pressure decreases when the velocity increases and vice versa.

Conclusion:
This concept could be applied to increase the downforce produced by the car. The fast-moving air above the car body creates a region of low pressure compared to the slow-moving air under the body. We know that air flows from regions of higher pressure to regions of low pressure. Hence, in turn downforce is produced.
Incorporation:
From our CFD testing we saw that the airflow around the car was already slow so we had to increase the speed on top of the car. To do this we had to create a laminar airflow at the top of the car which was achieved by creating fillets and having a streamlined design and ensuring there were no regions of turbulent air spaces.

RESEARCH & CONCEPT DEVELOPMENTS

3.1 /// NEWTON'S LAWS OF MOTION

Understanding Newton's laws of motion can have a significant impact on the speed of an F1 in school cars. From his three laws, the second and third law especially have a more significant impact on our cars. Hence it was important to understand these 2 laws in detail.

/// THE SECOND LAW

It states that when a net force acts on a body, the body accelerates in the direction of the net force.

The equation :
Force = mass x acceleration

Conclusion :
In F1 in schools, the thrust force given to all the cars is the same, since this force is derived from the CO2 canister being constant for all teams. We had to maximise the acceleration experienced by the car. To do this, the only variable of the equation under our control was mass which was inversely proportional to acceleration. Hence, the mass of the car had to be kept as low as possible.

Improvisation :

- To reduce the mass of the wings and wheels, we used Nylon 12 which has a relatively low density.
- Since the helmet and halo were not having a high stress impact, we used PLA as the material for its manufacturing which had one of the lowest densities.
- We ensured that the mass of the car model was a little under 50g, to take into account the 1-2 grams which could be added during the manufacturing stages.

/// THE THIRD LAW:

It states that every action has an equal and opposite action. This law gave us the indication to understand which forces were there and then draw up the action as a result of that force. This resulted in the creation of the free body diagram shown in the page before. Hence we were able to understand each of the forces affecting the car and modify the car in reference to them.

3.2 /// ASPECT RATIO

It is the ratio between the length of the wingspan squared and the surface area. From this it can be deduced that the aspect ratio is basically the wingspan divided by the chord.
Drag : A wing with higher aspect ratio will induce less drag force since the air will then travel along a longer surface area and will not meet as early as a wing with smaller aspect ratio.
Lift: Wings with higher aspect ratios experience less reduction in lift compared to those with lower aspect ratios because the wingtips are further from the main wing section.

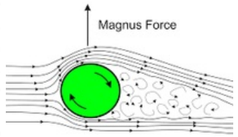
/// REAR WING DEVELOPMENT

We desired to have a rear wing with a higher aspect ratio to lower the drag force. To achieve this, we increased the rear wing span from 58mm to 60mm leaving a 5mm tolerance. Additionally we modified the rear wing chord from 17.5mm to 15mm. The CFD results on the wing supported the research.

	DESIGN 1	DESIGN 2
CHORD (MM)	17.5	15
DRAG (N)	0.043	0.031

3.3 /// MAGNUS EFFECT

Because of the regulations, we had to keep an empty space of 15 mm behind the wheels where we could not add anything. This creates a region behind the wheels which is completely empty and where nothing can be designed. This resulted in a region with turbulent airflow leading to drag and Magnus effect induced lift. Magnus effect refers to a phenomenon in which a force acts perpendicular to the direction of motion on a spinning object. In our case, the object is the wheels and the force is lift. This lift plays a crucial role in the performance of the car and to reduce it we made modifications to several car components.



Improvisation

- The endplates of the front wing were curved to redirect the airflow around the side of the car. This would reduce the air coming in contact with the wheel which in turn would reduce the Magnus effect induced lift.
- Since lift was being produced here, we had to counter this lift with downforce. To generate this downforce we made several modifications to the car. For example, increasing the angle of attack
- The angle of the sidepods play an important role in reducing the Magnus effect induced lift. From our virtual testing we saw that a slight upward angle toward the rear of the sidepods directed the air in such a way that it decreased the Magnus effect induced lift.

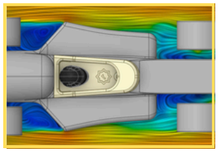
/// SIDEPOD DEVELOPMENT:

Design Concept 1 :
Our first side pod design concept was based on the concept of Coanda effect. We desired to ensure a laminar airflow at the sides of the car. This inspired us to create completely curved sidepods. It was also important for us to make sure that the incoming air from the front wheels was redirected away from the rearpods. We did this by adding a slight curve to the sidepods which tapered back to the original position. Additionally, the air which attached to the sidepods would create a low pressure surface over it compared to the air pressure of the surrounding leading to the air flowing from high pressure to low pressure along the sidepod surface of the car.



/// *Fig 3.1 | Initial Sidepod concept (Credit: Infinity Racing)*

Design Concept 2 :
Our second sidepod design involved all the properties of the previous design. However this time we ensured to keep in mind the magnus effect induced lift. To reduce this we created a section behind the front wheel which has 0 air velocity according to the CFD results. This indicated that the air speed was 0 behind the front wheel and there was no air speed behind the wheels showing there isn't any turbulent airflow behind the wheel which can produce drag. Additionally we wanted to make sure that the air was directed from the side of the rear wheel so we modified the curve of the first concept and removed the tapering curve at the end. We also ensured that none of the air from the front wheels entered the centre of the car from the side.



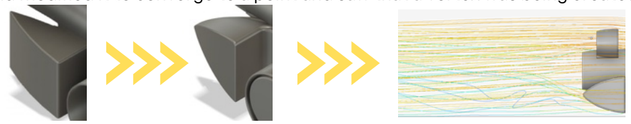
/// *Fig 3.2 | Image showing airflow around Sidepods (Credit: Infinity Racing)*

3.4 /// VORTEX CREATION

A vortex is when the air is directed in such a way that it swirls around an axis line. A a vortex behind the car helps reduce the drag faced by the car as it reduces the size of the wake(turbulent air) behind the car. Most of the concepts help increase the downforce at the front end of the car but for the rear end it can be done by the creation of a vortex behind the car.

/// REARPODS DEVELOPMENT

From our research we saw that to create a vortex we needed to shape the rear pods effectively. In our first concept we tried to converge the rear pod to a line, However we were not able to create a suitable vortex. Hence, we again researched and modified it to converge to a point and saw that a vortex was being created.



/// *Fig 3.3 | Development of Rearpods and CFD of Vortex (Credit: Infinity Racing)*

3.5 /// BODY WIDE VS NARROW

The width of the body plays a significant role in the car's performance. From our research we saw that it can affect the car's performance in parameters of : Aerodynamics, Stability and Weight Distribution. We designed a wide and a narrow body to asses them on these parameters and see which one is optimal for us.

PARAMETER	NARROW BODY	WIDE BODY
DESIGN & STATISTICS		
AERODYNAMICS	It had a streamlined shape which helped with aerodynamics. Since it had a comparatively less frontal surface area, it had a lower drag force.	The wide body had a comparatively higher drag due to the increased frontal area. But there is more downforce since the ends extend further from the centre.
STABILITY	Since the width of the body is less, it leads to a reduced distance between the left-right wheels reducing stability. The CoG was located closer to the centre line since most of the mass was in the centre.	It allows the use of a wider distance between the left-right wheels allowing a wider base for the car increasing the stability. From our testing, we saw that it helped reduce the tipping moment too.
WEIGHT DISTRIBUTION	We saw that most of the weight was concentrated around the centre line of the car. This led to the car not being balanced from the sides.	We noticed that the car was well balanced and the weight was distributed evenly throughout - front and sides. It helped increase the traction of the car.

KEY
DESIRABLE
UNDESIRABLE

We can see that the only weak point of having a wide body is the problem with drag. But there are several modifications which can be done to reduce drag in other components. Hence we chose to go with a wide body.

3.6 ///. FACTOR OF SAFETY

The factor of safety ratio is equal to the total amount of force that the car parts can handle divided by the amount of force that is actually experienced. It represents the extra strength and resilience that is built in the car to handle all of the unexpected forces and stresses that may occur during the race. By having a high factor of safety ratio, we were confident our car would not breakdown and could perform safely during the intense racing conditions experienced. It ensured our car was strong and reliable throughout the course of the races. We ensured that every one of our car components had a factor of safety ratio which was minimum 2. It was crucial to not compromise any other aspects of the part such as the weight, while strengthening it. The risk of any parts breaking or being damaged during the testing or racing phase was reduced to nearly zero thanks to our factor of safety ratio being of an acceptable number.

EVALUATION

Research and Development was seen to be the most important stage for the development of the car's performance. During this stage we evaluated concepts for different parts of the car and developed them according. We made modifications to the car's design according to scientific theories and principles related to aerodynamics and the car's performance. These modifications showed to have major improvements on the drag force faced by the car. Analysing the airflow around the car we were able to create a vortex behind the car which helped to produce downforce, additionally with these simulations we were able to decide the angle of inclination of the front wing and the curve of the sidepods. All of the modifications done were in accordance to the formula of development devised by the team which proved to create an efficient and easy method of carrying out modifications onto the car. Overall, all the modifications done to the car significantly enhanced its speed.
Improvement Actions : Addition of a curved angle to sidepods, rearpods converged to a point to create a vortex at the back of the car, the mass of the car was kept minimal to maximise acceleration, developed our design concepts for different parts of the car according to various scientific concepts and principles, modifications done to the car to decrease the drag and lift coefficient.

WHEEL DEVELOPMENT

4.1 /// MOMENT OF **INERTIA**

Our Understanding

The Moment of Inertia refers to an object's resistance to changes in its rotational medium. In our case, the object being considered under this were the wheels. The moment of inertia affected the amount of energy which was needed to set the wheels into motion. It was critical to understand the formula for calculating the moment of inertia so we could understand which factors we had in our control and reduce those to keep a low moment of inertia.

Formula

I = (m/2) x (R^2 + r^2)

Where I is the moment of inertia, m is the mass of the wheel, R is the outer radius of the wheel, and r is the inner radius of the wheel.

Conclusion and implementation

All the factors affecting the Moment of Inertia were in our control. The key takeaways from our understanding of the equation were:

- The outer diameter of the wheel should be minimum as to the regulations - we kept the outer diameter of the wheel at 28 mm.
- The width of the wheel should be minimum as to the regulations - keeping the width of the wheel 13mm for the front wheels and 17mm for the rear wheels
- The wheel should be as lightweight as possible but at the same time provide positive results in the stress simulations conducted on them.
- The thickness of the wheels should be kept minimum but it should not be too thick that the wheel would break or deform during the race.

4.2 /// MATERIAL **DETERMINATION**

From our research we saw that the properties of the different materials we had to research about were strength, weight, and flexibility, which can affect factors like acceleration and top speed. Material assessment criteria:

Density: Lower the density of the wheels, lower the weight and better is the performance as seen from the evidence above.

Durability: There is going to be high forces acting on the wheels during the race. The material should be strong enough to withstand the force without cracking, breaking or deforming. Stress test of 50N should be passed.

Heat Resistance: Heat can be produced due to the friction between the wheels and the track. The material should be able to withstand this heat without deforming.

Manufacturing Techniques: The appropriate method can help in manufacturing a precisely dimensioned wheel.

MATERIAL	DENSITY (g/cm^3)	STRENGTH (MPa)	HEAT RESISTANT	MANUFACTURING PROCESS
Nylon 12	1.02	50-80	Yes	Additive 3D Printing
ABS Plastic	1.04	40-50	Moderate	Additive 3D Printing
Carbon Fibre (CFRP)	1.80	1000-4000	Yes	Additive 3D Printing
Ketron PEEK 1000	1.32	90-100	Yes	CNC Machining

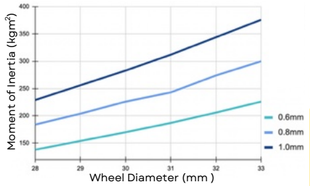
From the data in the table, we decided to use Nylon 12 as the material for the wheel as it passed all the criteria for the material of the wheel we were evaluating which have been spoken about above.

KEY

DESIRABLE | MODERATE | UNDESIRABLE

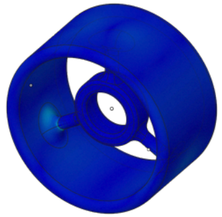
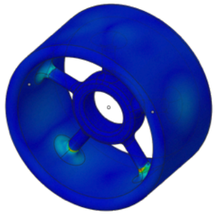
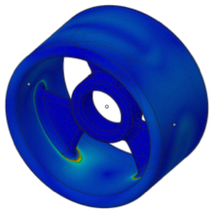
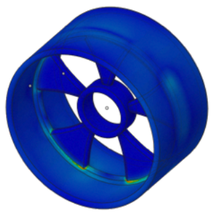
4.3 /// RADIUS AND **THICKNESS**

The outer diameter of the wheel and its thickness were directly proportional to the moment of inertia. When the outer diameter and its thickness were increased even the moment of inertia would increase which would resist the motion of the wheels. Hence, we designed our wheels to be as low as possible to be compatible with the regulation, keeping in mind the manufacturing considerations and tried to keep it as low as possible but at the same time pass the stress tests conducted.



/// Fig 4.1 | *Graph comparing Moment of Inertia & Wheel Diameter (Credit: Infinity Racing)*

4.4 /// WHEEL DESIGN **ANALYSIS**

NAME	3 ROD WHEEL	5 ROD WHEEL	3 SPOKE WHEEL	5 SPOKE WHEEL
DIAMETER (mm)	28	29	29	28
FEA ANALYSIS 10N | STRESS IMPACT				
RIM THICKNESS (mm)	1	1	0.8	0.8
MASS (g)	1.59	1.65	1.68	1.79
INERTIA (gmm^2)	1160	1204	1184	1286

The final wheel design had three spokes linking the central bearing housing to the outside rim of the wheel. Since testing and research showed that spokes considerably minimize rolling friction, this design was adopted.

The spokes of the design made the wheels very adaptable, thus it was simple to make minor adjustments wherever they were required without affecting the wheel's original design. It offered both a low moment of inertia as well as being able to handle high stress impoct.

4.5 /// ROLLING **FRICTION**

Our understanding

Rolling friction refers to the resistance faced when an object rolls over a surface.

Conclusions

- Speed : Some of the energy would be used for overcoming this resistance instead of increasing the speed and acceleration of the car. Increased rolling resistance would mean that the wheels were in firm contact with the track, hampering the rotation of the wheels.
- Energy efficiency : Rolling friction can cause energy losses in the car's system. Increased rolling friction would lead to energy losses in the form of heat, decreasing the energy efficiency of the system. Ensuring the rolling friction was minimised, more energy could be used to rotate the wheels faster and increase the energy efficiency system.
- Wheel wear : An increased rolling friction would mean that the tires would be in firm contact with the ground. The friction would cause the wheels to wear down after a while and hence decrease the consistency of the performance of the wheels. It was important to reduce the rolling friction to ensure an optimal grip with the track and increase the durability of the wheels.

Implementation

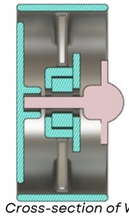
- Wheel Bearings : The use of high-quality full ceramic bearing to reduce the rolling resistance. We had to maintain the bearings with great care and lubricate if needed between the races. The bearing system of the car should be designed in the most efficient way possible.
- Wheel System Alignment : Ensure the wheels are well balanced and have a proper wheel design system with well aligned wheels to reduce unnecessary friction and ensure the smooth rotation of the wheels. We also increased the stiffness of the wheels so its deformation would decrease hence the rolling resistance.

4.6 /// BEARING **SYSTEM**

We modified our bearing integration for the INF 23. By suspending the bearing from the wheel while connecting it to the axil. Due to the bearings, we developed a design that would allow the wheel to roll with the least amount of rolling friction.

This task was accomplished by keeping the bearing within the wheel design with the aid of a housing like structure on one side, and a solid wall on the other. The axil will not spin as the car moves since it is attached to the body; as a result, only the wheel will revolve, reducing the friction between axil and the axil holes. This overall design has an extremely low rolling friction coefficient, thanks to the bearing integration.

The wheel is divided into 3 components, the axil, the wheel, and the cap. The cap is attached to the wheel from one side of the bearing housing, and the axil is attached to the bearing, which is trapped in the wheel from the other side. The axil also has an in built cap like structure to avoid airflow interfering with the wheel rotation.



/// Fig 4.2 | *Cross-section of Wheel System (Credit: Infinity Racing)*

EVALUATION

One of the most crucial components of our car are the wheels. Each aspect of it was evaluated using thorough research, which was then utilised to create a number of possible designs, each of which was tested numerous times before choosing which was to be implemented. We were aware that each material had advantages and disadvantages, and we used a comparison table and extensive research to ascertain which would be most advantageous. Graphs were used in the research on the relationship between radius and thickness, which revealed that the two variables were directly proportional.We designed a brand-new bearing system to be used for the World Finals after pinpointing the mistakes we had made in our National Finals system.

3D MODELLING

INFINITY RACING

3D MODELLING FOR MANUFACTURING

5.1 /// MODEL BLOCK **CONSTRAINTS**

The dimensions of the model block were fixed. Before getting the body manufactured, the body of the car model had to be compatible with the dimensions of the model block. The dimensions of the model block mainly included the total height (maximum = 50mm) , total length (maximum = 223mm) , total width (maximum = 65mm), CO2 cartridge position (29mm from the track and the position of the tether line guide(3mm either side from the middle of the body) . The position of the above features of the model block had to be the same as that of in the car model to ensure a precise manufacturing of the car body.

5.2 /// MATERIAL **CONSTRAINTS**

- *Foam Block:* The model block was made of polyurethane foam which has a low density making it a relatively weak material. If there was a thin extrusion of the body it would not be able to handle high force and could break off easily. Hence we ensured to keep a minimum wall thickness of 2.5mm throughout the body of the car.

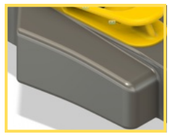
- *Nylon 12:* We used the Nylon 12 material for the manufacturing of the wheels and wings.It has a relatively high strength, so after our testing we saw that we needed to keep the minimum wall thickness for the wings and wheels at 0.5mm and wherever there was a relatively higher stress factor affecting that part keep the minimum thickness 0.8mm.

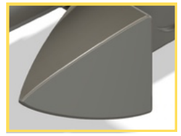
5.3 /// ASSEMBLY **CONSTRAINTS**

- *Nose Cone:* For the 2023 World Finals, a new challenge was added to the competition which was the nose cone replacement challenge. Hence it was important for us to have a quick and easy way of attaching the nose cone to the body. To incorporate this concept into our model, we thought of keeping a square shaped extrude at the front of the body which would mate with the similar cut extruded shape on the nose cone. To make this assembling process quick and easy we used the offset tool to create a 0.2 mm difference between the extrusion so that the nose cone would easily fit into the body and at the same time would be firmly held by the super glue. The 0.2 mm also kept in mind the space needed for the super glue.
- *Wheels:* The wheel system of our car included several different 3D printed components which had to be assembled together to create the wheel system of the car. It was essential that the bearing housing was not too tight which would affect the efficiency of the rotation of the bearings. Also the different wheel components should precisely attach to each other, to ensure that we kept a tolerance of 0.1 mm throughout the wheel system. This was made using the offset tool under the sketch section.

5.4 /// MACHINING **CONSTRAINTS**

- *Filleting:* For the manufacturing of the body, the CNC machine made the use of a ball nosed cutter which could not cut faces which were at a right angle or had a radius less than 2 mm. Hence, we ensured that no edges were at right angles and all of them had been rounded with the help of the fillet tool to at least a radius of 2 mm.



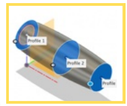


/// Fig 5.1 | Evidence of fillets used in car body (Credit: Infinity Racing)

USE OF ADVANCED MODELLING TOOLS

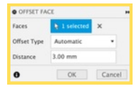
/// LOFT

The loft creates a transition between two or more sketch profiles or faces.
Application :
This tool was used to create the CO2 chamber of the body. By sketching multiple shapes on multiple planes, we were able to use the loft to create a shape that smoothly connected all the sketches to maintain the aerodynamic effect of the car.



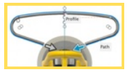
/// PRESS/PULL

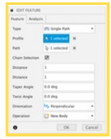
This tool modifies the selected geometry using offset, extrude and fillet commands. This is a quick and efficient way of doing relatively minor changes on the car than using other tools like extrude.
Application :
The CO2 chamber was created separate from the rest of the body. The press pull tool was used to efficiently join the two together to create a smooth transition both above and under the car.



/// SWEEP

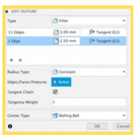
This tool allows us to extrude a profile along a selected path. The path as well as the profile can be sketched by us.
Application :
We used this tool to create the rear wing of the body. First we created a NACA 0015 aerofoil shape sketch using the circle, line and tangent constraint techniques. Then we created the path of the profile using the line and three point arc techniques under the sketch section. Both of these sketches in the respective places of the tool and created the rear wing. The sweep tool also offered us to directly create a new body out of the sweep done.





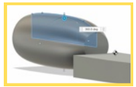
/// FILLET

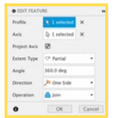
The fillet tool is used to round edges and faces. The rounding weight can be changed by adjusting the radius of the fillet. It also helps to avoid sharp edges which could be a safety risk. It is a more efficient and precise way than having to create sketches to round the edges.
Application :
Body - We used the fillet tool throughout the body at almost every edge, especially at the sidepods. It helps reduce the safety risk of the body and also make it more compatible with the CNC machine. The variable filleting option allowed us to efficiently change the radius of the fillet at the side pods so the design could be aerodynamically efficient as well as keep the strength of the sidepods in check.
Front wing and Nose cone - We applied a constant fillet of 2 mm, to all the exposed edges and faces when viewed from the front view to reduce the impact of the coanda effect.



/// REVOLVE

This tool can rotate a regular or irregular shape about an axis, with a selected number of degrees, creating a round body.
Application:
We used this tool to make the CO2 chamber of our body. Using sketching techniques, we created the shape of the body we wanted to revolve. This shape sketched was half of the overall desired body. By rotating the shape with the line of symmetry as the axis, the revolved body was acquired.





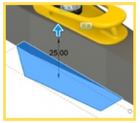
/// MOVE/COPY

This tool is used to move a body or component to another place or rotate it upon an axis. It can also be used to copy a body\component.
Application :
Used when assembling components together on fusion 360, to align the wings and the wheels to the appropriate position this tool was used. The copy feature of it was used to duplicate the wheels for the right and left sides.



/// EXTRUDE

The extrude tool is used to add or reduce depth to the selected faces, or sketches.
Application:
We used the extrude tool almost everywhere in our car. From creating 3D wheels out of sketches, to the front and rear wing to even the side and rear pods. Almost every sketch was extruded, whether it was to create a new body, join two bodies, or cut a body.



/// HOLE

This creates a hole based on selected values and specifications given by the designer.
Application:
Just as the shell, we used this tool to create a hole to house the CO2 cartridge in the body. This tool performed the task efficiently by creating a hole of diameter 18mm and height of 52mm.



/// COMBINE

This tool is used to join two or more parts of the body even if they are not physically touching.
Application:
This tool was used in many parts of the body. While designing parts, we came across multiple occasions where a modification to a part can affect other parts as a result. Hence, by designing them separately and then joining them with the combine tool, we were able to avoid this issue completely.



/// JOINT

The joint tool is used to define the relative position and motion between two components in an assembly.
Application :
The joint tool aided us in aligning the centres of the wheels to the centre of the axles, and see the rotational motion of the wheel on the axle. The cylindrical motion of the feature of the tool allowed us to see the rotational motion.



/// OFFSET

This tool creates the same sketch, feature or face at a specified distance or position from the original one. The shapes are based on the original geometry, but slightly resized or repositioned. It can also be used to create planes at a specified distance from a face.
Application :
We used the offset tool while making the connection between the nose cone and the body. We used this tool to create a 0.2 mm difference between the extrusion connection between the nose cone and body. The offset plane tool was used to create planes at distance from the YZ planes. These planes were used for making profiles for the loft tool used to create the part of the body around the CO2 chamber.



/// SHELL

This tool helps us in making the selected face hollow, and also allows us to create components with uniform thickness. Using this tool also proved to save time.
Application :
The lower part of the body had to be made hollow to reduce the mass of the car to increase the effect of the thrust force experienced by the car. With the aid of this tool we were able to make a precise and uniform 2mm wall thickness.



/// MIRROR

The mirror tool inverts the bodies/faces/features and copies them across a selected plane.
Application :
To ensure that the car was symmetrical and save time, we modelled structures on only one side of the car. Once the structure was ready, we mirrored that component across the midplane.



EVALUATION

3D Modelling was used for the manufacturing of the car and an in-depth analysis on manufacturing constraints like machining, material, assembly etc. To ensure precision and time efficiency during design and manufacturing, advanced 3D Modelling tools were used. These include fillets, shell, mirror, extrude, hole, combine, offset and many more.

COMPUTER AIDED ANALYSIS

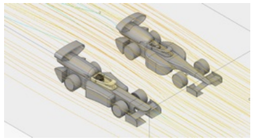
6.1 /// SOFTWARE **ANALYSIS**

The team's engineers carried out thorough research analysing the advantages and disadvantages of several softwares. From the research, it was seen the Ansys Discovery proved to be the optimal software for the team. The advantages clearly showed that this software had great potential like iterative testing, testing two models at the same time, etc.

ADVANTAGES	DISADVANTAGES
• Quick experimentation with geometry, materials, and conditions. • Real-time fluid flow simulation and visualisation. • Structural mechanical studies for stress, strain, and modal simulation. • Intuitive interface for easy control. • Iterative testing and design refinement. • Seamless integration with other Ansys products. • Two models can be tested simultaneously.	• Potential sacrifice of accuracy for speed. • Handling large and complex assemblies may be challenging. • Learning curve for proficiency with the software. • As the model is not designed on the same software, the software might have a problem in identifying some geometry

6.2 /// ADVANCED **TECHNIQUES**

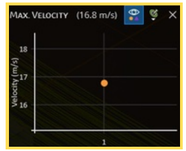
/// SIDE BY SIDE ANALYSIS



One of the critical advantages of this software was for us to compare two models at the same time. This was greatly beneficial for us as we were able to compare the results of two different models at the same time.
The main use of this tool, was when we had made a modification to the car design and wanted to check if it was enhancing the performance or not. By importing both the models, we could analyse the results of the earlier model and the model with the change to see if the modification was really making a difference, for example the change in the drag force or pressure areas in regions around the car. Another important use was when we were deciding to go for the wide body or the narrow body. We predicted to get a lower drag force with the narrow body sue to its streamlined shape, but the results were the opposite.

/// ITERATIVE GEOMETRY ANALYSIS

The software was able to generate a graph to show comparisons between two parameters : one geometry and the second was quantity like drag or lift force. This was used throughout the car to make minor modifications which would make drastic impacts on the performance for example increasing and lowering the drag force by 0.1N in some cases.



/// *Fig 6.1 | Point on graph of one of the iteratives tested (Credit: Infinity Racing)*

Application :
We used this method to determine the angle of incline of the front wing, we put 5 different angles varying by 1.5 degree each. From this we saw that 17 degree incline had a relatively low drag force as well as was capable of redirecting the airflow above the wheel.
We also tested different degrees of angles of attacks, having 5 values varying by 1 degree starting from 0 degrees, comparing the drag and lift force experienced by each. This was greatly helpful as then we were able to see which of them helped us to produce downforce and at the same time did not face a drag force higher than 0.2N. From the graph, we decided to go with the angle of attack as 1 degree.

6.3 /// CONDITIONS

In order to get accurate results, we had to set the physics of the enclosure to be the same as that of the race track.

- We kept the speed of the incoming air at 18m/s which was the average maximum speed the car would reach during the race.
- The density of the surrounding air was kept at the average air density which is 1.293 kg per metre cube.
- The wheel rotation was kept to be rotating at 1500 revolutions per second.

COMPUTATIONAL FLUID DYNAMICS

Computational Fluid Dynamics (CFD) is a branch of fluid dynamics and numerical analysis of fluid flow. This allowed us to visualise the airflow around the car, giving us vital numerical data which about the car's performance, for example the forces experienced by the car. We could also see the airflow streamlines around the car giving us an understanding of the regions of high/low pressure and also identify where there would be turbulent or laminar airflow which was very important.

6.4 /// METHODS

The model was imported into the discovery software, then using the external flow feature we created an enclosure around the model. The appropriate physics were set. Then we set the monitors which was mainly the lift and drag forces. Then all the necessary steps were taken and the simulation was ready to be solved. We ran the simulation and kept the refinement tab to full to get the highest quality mesh, so a more accurate result. Then we were able to toggle through the different views like velocity streamlines, pressure regions etc.

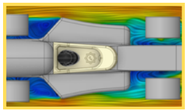


/// *Fig 6.2 | Evidence of Enclosure created (Credit: Infinity Racing)*

6.5 /// INTEGRATION **AND APPLICATION**

/// SIDEPODS

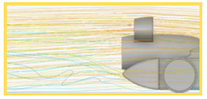
From the air speed zones, we saw that a region of turbulent airflow was being developed behind the front wheels. This was inducing lift due to Magnus effect. We made a modification which made the speed of the air behind the wheels almost 0, we then saw that this helped create laminar airflow behind the wheels. Another thing we wanted to make sure was that none of the air from the side pods was directed towards the rear wheels.



From the air speed zones, we saw that a region of turbulent airflow was being developed behind the front wheels. This was inducing lift due to Magnus effect. We made a modification which made the speed of the air behind the wheels almost 0, we then saw that this helped create laminar airflow behind the wheels. Another thing we wanted to make sure was that none of the air from the side pods was directed towards the rear wheels.

/// REARPODS

We ran velocity streamlines over the rear pods of the car and realised if we modified it such that it tapered back to a point, it creates an air vortex at the rear end of the car. We tested our theory by putting a vertical air streamline at the back of the rear pod. We saw that it led to the creation of a vortex at at the back of the car. Additionally, from our research we saw that is helps generate downforce. This was proved by the reduction in the lift force of the car by 0.1N which was a significant decrease relatively to other modifications.



/// *Fig 6.3 | Evidence of vortex created on rear end of car (Credit: Infinity Racing)*

/// FRONT WING

The angle of incline of the front wing was one of the most important factors influencing the drag force of the car. We tested its angle of incline to see what was the lowest angle we could have at the same time, all the air was being redirected above the wheels. The lowest angle was required to ensure that the drag force was kept minimum. This was done by using vertical velocity streamlines to see the airflow around the front wing. We used the horizontal streamlines to check whether the endplates of the front wing were deflecting the air appropriately, which was away from the front of the wheels. In the first simulation, we saw that the curve was not enough so we increased it by 2 degrees and then saw positive results.



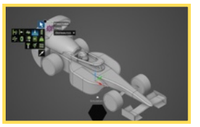
/// *Fig 6.3 | Deflection of air by front wing (Credit: Infinity Racing)*

FINITE ELEMENT ANALYSIS

Finite Element Analysis (FEA) is a computerized simulation approach used to dissect complicated structures into smaller, simpler components known as finite elements in order to analyze how they behave. A body's response to forces, vibrations, heat, fluid movement, and other physical circumstances is predicted using FEA. They were carried out on Ansys under predetermined conditions like drag and lift coefficients. To get the most accurate results, it was necessary to provide everything, including the actual materials of the components and the limitations.

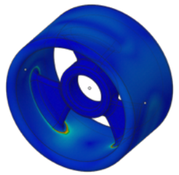
/// FORCE TESTS

Since no other parts of the car are blocking them in any way, the front and rear wings are subject to the most force when the car moves. In order to analyze how the wings would respond, we conducted several force tests on them both separately and in conjunction with the automobile. Until the ideal combination was found, this was repeated with adjustments to the materials and the design. We were able to track the development of the designs as we made adjustments by gradually increasing the forces encountered.



/// STRESS TESTS

Due to the extreme rotations, the wheel walls experience tremendous stress, which can lead to cracking or deformation, which would go against the safety regulations and result in a slower overall speed of our car. To combat this issue, we thickened the rim of the wheels, and integrated spokes inside the rim of the wheels. This increased the structural integrity, while also reducing rolling friction of the car. By pinpointing the exact positioning of the extreme stress, we were able to make changes to the design for the wheel to withstand the pressure applied.



/// TEMPERATURE TESTS

Since our car travels at extremely fast speeds during the race, the temperature of the car surface will significantly increase not only because of the high kinetic energy in the car, but also the friction caused by air as our car travels through it. To combat this issue and make sure it didn't hamper the structural integrity of our car, we tested the car virtually with several different temperature levels, gradually increasing the temperature from low to high to see how our car would perform in such situations. Making material, or design changes in our car allowed our car to have a strong base, which would not be affected significantly by the high temperatures experienced during the races.



EVALUATION

Computer Aided Analysis is a critical stage in the design and engineering process to visually analyse the car in a simulated environment. Ansys was used as the software for CAA as its mentioned advantages outweighed the disadvantages. Furthermore, advanced techniques such as Side by Side analysis and Iterative Geometry analysis were conducted and applied to improve the car's design. Computation fluid dynamic tests and Finite Element Analysis was also ocnducted on the car's components to ensure a high performing car.
Improvement Action: Angle of incline was increased on the front wing; A curve was added to the sidepods and the rearpods were converged to a point to improve the car.

7.1 /// MANUFACTURING **GOALS**

Accuracy: Our entire manufacturing process including the CNC machining of our body and the printing of our car parts was supposed to be as accurate as possible with a tolerance value of +- 0.2 mm. This was to keep our goal of designing and manufacturing a car of high performance and durability.
Machine Dependent: A goal was to keep our entire manufacturing process dependent on machining and minimising the use of hands to avoid errors and keeping our accuracy to the maximum.
Safety: Safety is an important factor and keeping it as a manufacturing goal was critical. It is important that the manufacturing process does not cause physical harm to anyone.
Regulation Compliant: Manufacturing a car that strictly complies with the set regulations is critical to gain valuable points in the competition. With high accuracy, we also aim to keep a 100% regulation compliant car.
Time: Entire manufacturing process should be time efficient to avoid delay in later processes like assembly and finishing. Selection of local outsourcing partners for manufacturing is critical to keep the process time efficient.

UNDERSTANDING THE TECHNOLOGY

7.2 /// UNDERSTANDING **THE MACHINE**

Before starting with the CAM process, our team wanted to understand the specs of the CNC machine we could be using so that we could use it to its full potential. We used Denford's VMC1300 Pro CNC machine for our car manufacturing. We were able to obtain the relevant information needed about the machine from Denford's website.



TRAVEL X-AXIS	375 mm
TRAVEL Y-AXIS	160 mm
TRAVEL Z-AXIS	235 mm
MAXIMUM SPINDLE SPEED	6000 rpm
MAXIMUM FEED RATE	5000 mm/min

7.3 /// UNDERSTANDING **THE SOFTWARE**

CAM software is essential when manufacturing tools are being used, they aid us in manufacturing a precise model. In our case we used the CAM software for generating the G-Code for our CNC machine. We used Denford's QuickCAM pro software to generate our codes. One of the most significant advantages of the software was that it was tailored to manufacture F1 in school cars, which made the process much smoother, like the software already had its fixture and stock model installed due to which we did not have to manually set our own boundaries. Secondly, the interface of this software is extremely easy to understand and it is not very complicated to run the necessary processes on it. Moreover, the CNC machine we were using was the Denford MRC 40, and this software was the most compatible with the machine. We were able to modify and adjust several parameters like the spindle rate, feed rate, cut width, cut depth according to our need. These features gave us complete control over the CNC process, enabling us to use the CNC machine to its full potential.

CNC MILLING PROCESSES

7.4 /// MODEL PREPERATION **(BEFORE CAM)**

- Virtual inspection of the CAD model to check whether it is compliant with the dimensions of the model block and not part of it goes outside the model block. Also the position of the CO2 chamber is correct as well as the position of the tether line guide was taken into account.
- All the edges of the body should have a minimum fillet of 1.5mm and minimum thickness of any part of the body should be at least 2 mm since it is a rather weak material.
- We exported a .stl file of the car body model.

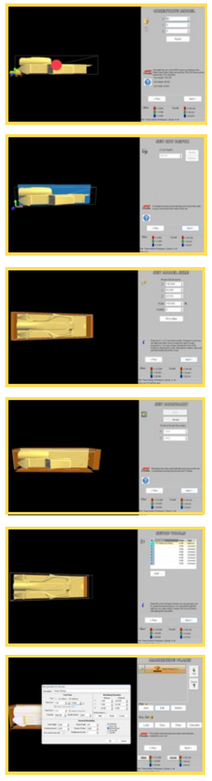
7.5 /// **PLANNING**

Before generating the codes for the CNC machine, we took a session with our mentors to understand how the CNC machine works and discussed with them the best techniques which should be used for the manufacturing of our car. We also then came to know that it would benefit us if we first carried out the process for the bottom side, then right hand side then left hand side. If there was a difference for the particular step it has been mentioned in that step.

7.6 /// QUICKCAM PRO **PROCESSES**

The following process outlines the steps undertaken for the generation of the code for the milling of the right hand side of the body. The left hand code is automatically generated once the left hand code is generated. We had to create a separate code for the bottom body which was a little different than the other two.

1. Import the file - After opening the software,using the car wizard feature we imported the car body's .stl file into the software.
2. Orientation - The model was oriented in such a way that the red arrow was pointing. The red arrow shows the side where the cutting phase will occur. In our case, this was the RHS of the model. The arrow was positioned on the correct side by changing the X,Y,Z values on the right hand side of the software.
3. Set Cut Depth - Depth cut refers to how deep down the drill is allowed to go. In our case, when we were generating the RHS code, the cut depth was automatically set for the RHS side and we did not change this as it was the correct value which was 38mm. For the bottom side we had to fix the value for the cut depth, we set it at 39mm which was a little above the sidepods.
4. Set Billet Size - The billet size refers to the size of the block on which the process is happening. For us it was the F1 in schools foam model block. The dimension of the billet was kept as follows, x=223, y=65, z=50.
5. Set Model Size - We ensured that the model size was kept at 100% and also that the dimensions for the model size matched the dimensions given in the bottom right hand corner.
6. Set Model Position - The position of the model block was sent back until the rear surface of the car was at the same surface as the rear face of the billet which is the model block.
7. Set Boundary - This refers to whether the whole billet was going to be cut or the part of the billet in which there was the model. For us, the size of the billet and model were similar so we selected the boundary as the billet.
8. Setup Tools - We made sure that the tool was a 6.35mm ball nose cutter.
9. Machining Plans - We used the Raster machining plan for the machining plan of the model. The Step over setting refers to the model's finish quality. Lower the step over more the time it takes for machining but it is more accurate. Hence we kept a step over of 18% which ensured a smooth finish over the car body with minimal line marks. After adding the step over value, the other 2 important values the team had to decide upon are the spindle speed and the feedrate. The feed rate refers to the speed with which the spindle advances into the model. We saw that the lower the feed rate the more accurate the machining. Hence we kept it at 3000 mm/min. The spindle speed refers to the angular speed of the drill. Once again to achieve an accurate model we used a relatively low speed which is 23000 RPM. We tested several combinations and the above combination served to be the best.



10. Toolpath Simulations - Once the construction plan was set, we were able to run a toolpath simulation. We were able to get an idea of how the machined model would look like in real life. WIth the help of this simulation we were able to test several combinations and compare them to check which of them gave the most accurate and smooth finish.
11. Set Datum Position : The last step was to set the datum position as the centre of the CO2 chamber. Then we exported the fnc file which is the G-Code. The code for the LHS was automatically generated but the process had to be repeated for the bottom of the body.



/// Fig 7.1 | Images of important steps of the process (Credit: Infinity Racing)

7.7 /// MACHINE **PREPARATION**

1.Machine table cleaned using vacuum pads
2.Foam block placed on top of the material
3.Securing cutting tools by tightening bolts on machine tool holder
4.Machines drill moved to side of material for correct positioning for machining
5. Calibrating the machine so that instructions are followed and is ready to begin manufacturing

7.8 /// POST **PROCESSING**

On removing the car from the machine we immediately cut off all of the extra bits that were attached to the car before entering the final manufacturing phase. After this, these ends were sanded in order to ensure all of their surfaces were smooth.

GENERAL

7.9 /// SAFETY AND QUALITY **CONTROL**

RISK	SAFETY MEASURE	RISK IMPACT	LIKELIHOOD	QUALITY CONTROL
During manufacturing and sanding of car components, dust particles may be produced which can be inhaled.	Safety goggles and ear plugs to be worn at all times and immediately vacuuming the component after the process is complete.	HIGH	POSSIBLE	Prioritised top-notch printing techniques that result in smooth surfaces, minimising the sanding need, and always wearing the necessary gear for protection .

7.10 /// MACHINING **TOLERANCE**

While developing our cars, we used a standard tolerance which would ensure top notch quality of printing of parts with minimum requirement for finishing by hand. Our minimum tolerance was 0.1mm while our maximum tolerance was 0.2mm. In our car designs we left a bit of margin of space incase of any manufacturing issues.
Varied tolerances were employed for specific car sections, addressing concerns like paint accumulation and adherence to regulations. In designated exclusion zones, such as the wheel areas, a 1mm tolerance was adopted to accommodate additive manufacturing processes.
The CNC milling machine, with ±0.1mm tolerances, could be influenced by heat-induced expansion. To counter this, we prioritised concise milling plans to minimise any tolerance shifts due to heat.
To account for milling inaccuracies, a 0.2mm finishing amount was introduced, ensuring the original design integrity and reducing the need for extensive hand finishing.

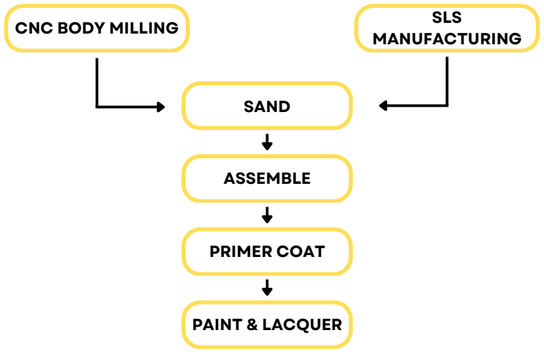
EVALUATION

To ensure a smooth CAM & CNC process, manufacturing goals were set to enhance the performance of the car. Understanding the overall technology including the CNC machine and the software was critical to perfect the process. The pre-printing and post-printing processes was also analysed and understood including the model preparation and the QuickCam Pro stages. Moreover, safety and quality control was critical to prevent possible harmful risks.
Improvement Actions: In order to achieve precision and quality in the printing and post-printing process, a reduced feed rate and slower spindle speed would help improve the precision of the manufacturing stage.

MANUFACTURING & ASSEMBLY

8.1 /// PROCESS **OVERVIEW**

The entire manufacturing and assembly process were subdivided into three stages;
Materialising, finishing, and assembly.



8.2 /// MATERIALISING

/// PRINTING

The next step of our process was to synthesise our car. We designed the virtual model of the car using CAD technologies and made it physical using CAM. We utilised a 3D printer to print the different parts of the car separately. For parts such as the Nose Cone/Front wing, Wheels and the Rear Wing, Nylon 12 was used as the material due to its high strength and it is relatively lightweight compared to other analysed materials. For the housing and the halo, PLA was used for its lightweight. The entire printing process was conducted by our partners, Forge 3D and Wol 3D, but was also monitored by our team.

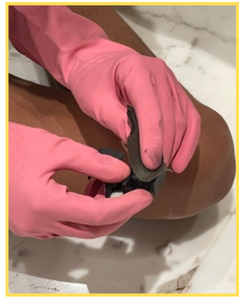
/// POST - PRINTING PROCESS

Once the parts were received after 3D Printing, all parts were brushed to remove excess of the Nylon 12 material which was covered on the parts in powder form. This concluded the post printing process and the parts were kept for 24 hours to cool off. The finishing process was to be carried out now.

8.3 /// FINISHING

/// SANDING

Following the CNC and 3D printing of the body and the parts, the surfaces were not smooth with excess material in powder form. Hence, the body and parts were initially sanded using a 400-grit sandpaper. A low-grit sandpaper was first used to remove the powder form and to prevent straight away damage to the parts or the body. Then, a 600-grit sandpaper was used to finally sand off the parts and the body to make the surface smooth.



/// Fig 8.1 | Image of the Sanding process (Credit: Infinity Racing)

/// ASSEMBLY

After the sanding of each component and the body, the components were initially placed on the body using a 2D sketch diagram we created showing the placements of all the components. A dry run was conducted while doing this to ensure all components were of the correct size and were fitting properly to avoid any gaps.

Hot glue was used as the agent to assemble the entire car. After the assembly of all components to the body, the entire car was left on our initial 2D sketch diagram for 1 hour. This was to ensure no movement was taking place among the components from the body and if all parts were completely aligned. Lastly, the parts were gently pressed to the body to verify the assembly process.



/// Fig 8.2 | Image of the Assembly process (Credit: Infinity Racing)

/// SEALING

After assembling the car, we noticed spaces and gaps between the components and the body. This was prominently seen between the nose cone and body connection. To avoid this, we used cork to seal gaps and spaces. This conclusively increased the strength of all the connections. Lastly, another round of sanding was conducted using 400-grit sandpaper along the parts where the gaps were filled to level the material, concluding the sealing process.

/// PRIMER

Two coats of primer were applied through the entire car to ensure a smooth surface was created on the car for an easier painting process. There was an interval between the two coats to let the primer dry. After each coat of primer, a light 400-grit sandpaper was again utilised which ultimately made the surface extremely smooth for the latter finishing processes.

/// APPLICATION OF PAINT AND LACQUER

2 coats of automotive paint were applied to the car once the primer had been sanded and completely dried. This created the car's livery. Masking tape was used to cover certain parts where a different colour of paint was required. The paint coats were applied in time intervals to ensure the previous coat had dried. After the paint layers had dried up, lacquer was applied to the car to create an extremely smooth layer and give the car a glossy finish. 2 coats of lacquer was sufficient to give the car its smooth finish. These were applied once before the pasting of the decals and one after. The process of applying paint and lacquer was outsourced to our partner, House of Polish. However, the entire process was closely monitored by our team members.



/// Fig 8.3 | Image of the Painting process (Credit: Infinity Racing)

/// APPLICATION OF DECALS

To showcase our sponsor's logos, waterslide decals were used on our car. The decals were soaked in water for 30 seconds and were then placed on the respective sponsor locations and the decals slid onto the car. After 1 hour, another coat of lacquer was applied to conclude the process by giving the car a smooth and glossy surface.

8.4 /// QUALITY & **SAFETY**

STAGE	QUALITY CHECK	HEALTH & SAFETY MEASURES
SLS 3D Printing Car Components	▸ Strength tests to ensure that the machine isn't malfunctioning and misprinting parts. ▸ Strict inspection and failing policy when problems occur.	▸ N95 masks used to filter all airborne particles ▸ Members to wear PPE including safety goggles and gloves ▸ Ear buds to be worn during printing
CNC MILLING	▸ Milling at lower speeds ▸ Ensuring drill bit is undamaged	▸ Ear buds to be worn during milling ▸ Members to wear PPE including safety goggles and gloves ▸ Clear chips and debris from milling area to ensure they do not interfere with the process
CAR ASSEMBLY	▸ Before assembly starts, ensure no components are damage and defect free through thorough inspections. ▸ Verify that all parts are properly fit in place ▸ Car placed on a flat surface ▸ Inspection using measurement guides used in order to find best performing car to be used on track	▸ Clean and flat surface for car to be assembled on ▸ Gloves worn to make sure sharp components do not cause cuts or any other form of injury
SANDING	▸ Visual examination done to identify areas which require sanding ▸ Proper grit selection for specific areas ▸ Sanding in a cautious and suitable pace in order to avoid scratches ▸ Thoroughly clean the car after sanding is complete to remove all residue	▸ Mask to be worn to prevent inhalation of airborne particles ▸ Goggles to be worn to prevent entry of dust and particles into the eye
PRIMER COATING	▸ Ensure underlying component is completely dry so that surface is smooth ▸ Applied in suitable circumstances to ensure coat is applied evenly	▸ Well ventilated surroundings ▸ Gloves to prevent skin contact with primer
PAINT AND LACQUER APPLICATION	▸ Applied in suitable circumstances to ensure coat is applied evenly ▸ Previous coating allowed to dry for adequate amount of time to ensure smooth surface ▸ Thorough visual inspection ▸ Maintain sufficient drying time between layer applications	▸ Well ventilated surroundings ▸ Gloves to prevent skin contact with primer ▸ Wearing full sleaved apparel to ensure that none of the paint came in contact with the skin ▸ Wearing a mask while painting
DECAL APPLICATION	▸ Recheck dimensions of decals ▸ Recheck placements with rendered images ▸ Ensure F1 in Schools decal is clearly visible through thorough visual inspection	N/A

8.5 /// OUTSOURCING **JUSTIFICATION**

PART	JUSTIFICATION
3D PRINTED COMPONENTS	We formed a partership with a local professional 3D printing company to 3D print out car parts. The components were printed in Nylon 12 material as it had lower density than carbon fibre, possessed a satisfactory amount of strength and was heat resistant as well. This allowed us to create components which were sturdy enough to withstand forces experienced during racing while not compromising on the total weight of the car thanks to the light weight of the material. SLS 3D printing is a manufacturing process that utilises powdered plastic, where thin layers of the material are fused together. This technique enables the creation intricate and detailed parts without the need for support material, resulting in flawless components with no visible layer lines.
CAR FINISHING	For the World Finals, we wanted to have a professional and smooth finish of the car, which we weren't able to do in the National Finals. The car's paint job and finishing was outsourced but closely monitored by our team members. We outsourced it to 'House of Polish', located in Delhi for the painting of the car after it was successfully printed. Although it was not in the same city, we managed to travel there and understand the method and techniques of doing it. They used efficient methods to paint the car such as applying primer, automotive paint and lacquer.
BEARINGS	Regarding our bearings, we trusted our sponsor Abec357 who have been on this journey with us since the regional finals and hence outsourced it to them. We used the full ceramic bearings which we had purchased at a great discount thanks to our partnership with the supplier. We did not intend on purchasing bearings locally as we had previously tested them only to see that they performed at a scale nowhere near the Abec357 full ceramic bearings, and hence believed this was a department which we could not afford to.

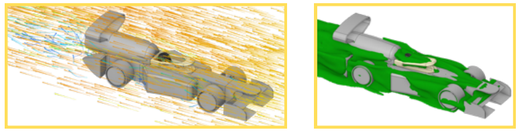
EVALUATION

Thorough research into the field of 3D printing materials was conducted which was a pivotal piece of data we needed to collect. The car's surface finishing procedure required meticulous execution in order to ensure utmost quality, subsequently bolstering the car's overall performance. Diligent quality checks at specific times was imperative, and stringint safety protocols were exercise at all times through these processes. All of these measures collectively emphasised our commitment to achieving excellence in both design, as well as manufacturing .

VIRTUAL TESTING

9.1 /// CFD **ANALYSIS**

Goal: The CFD simulations were mainly conducted to understand the impact of the aerodynamic forces like drag and lift on the car's performance. We wanted to make sure that the flow of air around the car was optimised. It would help us to understand whether the design of the front wing, rear wing and side pods were appropriate, for example the angle of incline for the front wing.
Method: We imported the car geometry onto the Ansys Discovery software. We set up the simulation with the suitable parameters and monitors and then started the simulation. We kept the wind speed at 13.4 m/s to check the impact of the maximum air speed. We looked into the air displacement, drag and lift for each of the components.
Outcome: After this we made major modifications to the designs of the front wing and rear pods. Visualising the airflow around these parts, drag and lift significantly benefited us. We were able to reduce the drag coefficient from 0.47 to 0.29.



/// *Fig 9.1 | CFD Tests run on car (Credit: Infinity Racing)*

9.2 /// FINITE ELEMENT **ANALYSIS**

Goal: To ensure that the car components would be able to handle the force of the air and were not deforming because of the stress. To identify the potential areas of weak structures and modify them appropriately. To see if the safety factor we
Method: We used the structural feature on the Ansys Discovery Software to see if the component was able to handle the stress. We imported the geometry onto the software and set the suitable material for it. Then we put the maximum force which was possible for that component and checked whether there was a light blue colour over the component indicating it was able to handle the stress.
Outcome: We were able to find out the optimal wheel thickness which was at 1 mm(shown on page 2). First we had kept the wall thickness of the body at 2 mm but we noticed that it was not able handle higher loads so we increased it to 3 mm. We saw that the front wing was bending at higher speeds so we increased its thickness. The rear wing thickness was kept at 1 mm but it was deforming at high loads so we increased the minimum thickness of it to 1.5 mm. We were able to see that the safety factor we kept was sufficient.

9.3 /// CENTRE OF GRAVITY **CALCULATION**

Goal: We wanted to find out the position of the CoG in the fully assembled car model to see if the position of the CoG was similar to the one desired.
Method: We assembled the full car together on the Fusion 360. Using the centre of gravity tool under the inspect section on the software we were able to determine the position of the CoG of the fully assembled car.
Outcome: We were able to see the CoG for each of these cars. To get the point aligned to the correct position we made modifications like increasing the mass at the front or back to move the CoG to the back or front to achieve an optimal location for the CoG.

9.4 /// COMPUTER AIDED MASS **CALCULATION**

Goal: We wanted to design a car which had a mass of approximately 44g according to the virtual model, leaving 6 grams which would be required for the finishing of the car like the layer of paint. We also set an approximate mass goal for each of the car components.
Method: We used the physical material tool on the Fusion 360 software, to set the desired material for each of the different components of the car. Some of the materials were not available on the software so we had to manually enter the density of the material, for example we manually entered the density of polyurethane since the exact density of the material was not available. Then we went into the properties of the car and were able to see the mass of the car.
Outcome: With the help of this tool we were able to see which of the different materials we could use and test different combinations of them to see which one worked the best. With the help of this tool we were able to set the virtual mass of the car to 44g leaving a tolerance of 6g for the finishing.

PHYSICAL TESTING

9.5 /// REGULATION COMPLIANCE **TEST**

Goal: To achieve a car design which is compliant with all the regulations, losing 0 points as penalties in the car Scrutineering points.
Method: To do this we used the sketch dimension tool on Fusion 360, to check the distance between two car components or two points. For a few of the regulations we had to use the create sketch tool to insert a line and measure its length to check the dimensions.
Outcome: We were able to rectify any problems with the technical regulations, before the car was manufactured.



/// *Fig 9.2 | Image of measuring specifications for regulation compliance test (Credit: Infinity Racing)*

9.6 /// SMOKE **TEST**

Goal: To view and analyse the airflow around the car and identify components for improvement.
Method: Place it in the smoke testing tunnel and place incense sticks to simulate smoke. Put a fan behind incense sticks to direct airflow towards the car.
Outcome: The incense sticks created small black marks depicting where the wind primarily hits the car. The components with the darkest spots were considered to be improvised for the final car development. We noticed dark spots on the top areas of our front wheel and on our side pods for initial prototypes which were then reduced significantly for our final car. Smoke testing was a crucial test to determine airflow around the newly added halo.

9.7 /// WIND TUNNEL **TESTING**

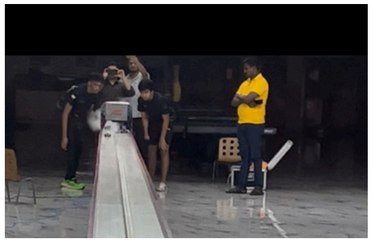
Goal: To optimise the car's performance in terms of aerodynamics and verify our CFD analysis. Incase CFD results do not match with Wind Tunnel results, find out reason and improvisation method for the same.
Method: The car was placed in the Wind tunnel and the wind speed was set at 14 m/s. This is chosen as the average speed of the car. We put in Dry Ice to simulate the wind and the entire testing was recorded to view results multiple times.
Outcome: Overall, the Wind Tunnel results matched with our CFD results. The windflow around a car was as expected, going around and above the wheels due to the correct functioning of our front wing and side pods. The air was displaced by the rear wing as expected.

9.8 /// BEARING **TEST**

Goal: Analyse and evaluate various bearings' rotational performance, in order to make informed decisions in selecting the most efficient and smooth bearings for optimal performance.
Method: To test the bearings we used a drill which was attached to the wheel following which we removed the drill and then timed how long it took for them to come to a stop. Tachometer was used to make all the wheels spin at exactly 1300 rpm and each bearing spun 5 times using which average can be calculated.
Outcome: The bearings' test helped us to determine which bearings were performing the best i.e. has the smoothest rotation. We initially were planning to use hybrid ceramics, but after the test noticed that full ceramic bearings were performing better and hence decided to use them instead.

9.9 /// TRACK **TESTING**

Goal: To evaluate the performance of the car, check for durability and reliability of it in race conditions and find the highest performing car amongst those which we had printed.
Method: Conducted track testing sessions during which it will be checked which parts are most likely to incur more damage after every run. All canisters used to store in the cooler prior to test runs at the same temperatures.
Outcome: Through this method we found out that our axles were made thicker as we noticed them snap multiple times during track runs.



/// *Fig 9.3 | Image of our car during track testing (Credit: Infinity Racing)*

9.10 /// REAL LIFE MASS **TEST**

Goal: Real life mass should be similar or equal to expected mass from virtual mass tests. Mass of the car after assembly and finishing should be as close to 50g as possible and should not exceed 51g.
Method: After assembly and finishing of both cars, cars should be placed on digital weighing balance and mass of both cars should be noted.
Outcome: Our real life mass was extremely close to our expected mass as per our virtual mass testing. Our goals were achieved where both the car's mass was the same and in the range provided. The mass of both cars were approximately 48.6g before painting.

9.11 /// TETHERLINE GUIDE SAFETY **TESTING**

Goal: Ensure that tether line guides on the car meet safety standards and comply with the regulations given in Technical Regulations.
Method: Post assembly of tether guide, weights of about 200g hung from them in order to find out whether it can withstand the corresponding forces and race safely on track.
Outcome: The tether guide complied with the regulations and that our FEA predictions were correct.

EVALUATION

It was imperative for our team to conudct a number of rigorous tests on the car. We undertook a wide range of physical tests, carried out on the actual car, as well as employing the usage of several virutal softwares to conduct simulations on the design. The insights gained from these tests were instrumental in scrutinising our car's design and identifying regions for improvement in order to enchance pace while not compromising durability. All test results were analysed in an adept manner, thus clearly determining the modifications required to better the performance of the car. This iterative process of testing played a pivotal role in our pursuit of success on the design aspect of the competition.

CAR CONCEPT DEVELOPMENT

10.1 /// NATIONAL FINALS **CAR**

Drag Coeffiecient : 0.35
Lift Coefficient : 0.27
Mass of model : 52g



Development Goals from Qualifiers Car :

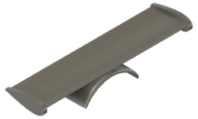
- The front wing should redirect the air above the wheels
- Do more research on the wheel system and make it more effective
- Creation of downforce with the help of the rear wing

/// Development of Goal 1

It was seen that in the qualifiers front wing, the air was just being cut, relatively less air was being redirected around the wheels and quite a significant amount of the air was hitting the middle of the wheels. Hence for the national finals we went ahead in incorporating aerofoils which not only helped in the redirection of the air but also helped in the generation of downforce at the frontal end. This was confirmed by the CFD result comparison between the cars. However, we were not extremely happy with the aerofoil front wings since the airflow around the wing was yet not optimal and its mass was relatively high too.

/// Development of Goal 3

We used the NACA 0015 shape for the rear wing of nationals car. Not only did this shape have a low drag coefficient but also can be used to generate downforce.

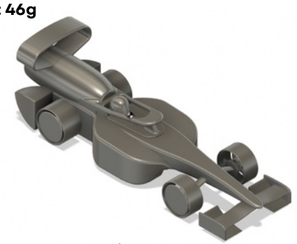


EVALUATION

The modifications made to car showed clear improvements in the car's design as the track time time decreased by 0.2 seconds from the qualifier's design. This not only showed the car's improvement but also proved that the team's method of developing the car and concepts were appropriate. The car severely lacked an efficient bearing system as the bearings were vibrating about their position vigorously. The front wing was aerodynamically inefficient as a lot of the air was being blocked and obstructed due to this components, hence effecting the overall airflow around the car. Overall Mass of the car was also far more than what our target was, 59 grams, and this greatly affected the pace of the car during racing.

10.2 /// PROTOTYPE **1**

Drag Coeffiecient : 0.31
Lift Coefficient : 0.24
Mass of model : 46g



Development Goals from Nationals Car :

- Enhance the airflow around the front wing keeping its mass in mind
- Do more research on the wheel system and make it more effective
- Incorporation of halo into car design

/// Development of Goal 1

From our research we saw that the best way to redirect the airflow around the front wheels was to use a slope like front wing. Not only is this front wing effective but also the mass of it was comparatively less. But from the CFD simulation we saw that some of the air was yet hitting the top of the wheel. To reduce this we put a blade a on top of the front wing, which reduced the air hitting the front wing and redirected it above the front wheels. By changing the angle of attach of this blade we were also able to generate downforce. Our findings were proved by the CFD results.



/// Development of Goal 2

Since the qualifiers stage, the wheel system of the car had not been updated and worked on properly. We had quite an innovative idea for the wheel system which would reduce the rolling friction greatly. This system has been described in detail in section 4.5.

/// Development of Goal 3

The newly introduced halo needed to be studied in depth and the optimal position of it in the car could greatly enhance its performance. We ran CFD simulations on the halo model specifically and saw that it was splitting the air into two directions. We saw the body of an actual F1 car and saw that the shape behind the halo. We mimicked a similar shape into out prototype and saw that the air was smoothly flowing from the sides of the body behind the halo.

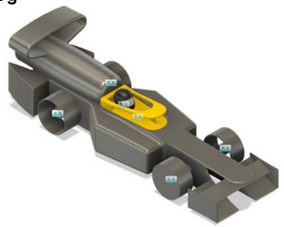


EVALUATION

The nose cone structure was far too thin, which made it fragile, and would suffer breakage on impact with end of racetrack. Weight distribution of car was not ideal, hence causing imbalance which would lead to more weight being applied on a specific region of the car. A large amount of turbulent air was being generated due to the sidepods which had a massive effect on the overall performance of the car.

10.3 /// PROTOTYPE **2**

Drag Coeffiecient : 0.30
Lift Coefficient : 0.2
Mass of model : 47g



Development Goals from Prototype 1 :

- Try having a wide body which would enhance the weight distribution
- Develop the sidepods to reduce the turbulent airflow behind the front wheels
- Minor modifications to car to enhance performance

/// Development of Goal 1

From our earlier prototype we saw that most of the mass was being concentrated in the centre of the car. The sides of the car were not getting enough weight. This lead to the decrease in traction between the wheels and the track. We were able to see the traction only when we physically manufactured the car so it was a relatively difficult to get the appropriate traction since we could not keep manufacturing the prototypes to test the right traction.

/// Development of Goal 2

From our research we saw that there was a region of turbulent right behind the front wheel. This was inducing lift due to Magnus effect, increasing the lift coefficient. So we created a design which significantly decreased the amount of the behind the front wheel. The evidence of this can be seen in section 4.3.

/// Development of Goal 3

We saw that the there was a tipping moment being created at the rear end of the car. By increasing the angle of attack by 1 degree we saw that we were able to generate downforce at the rear wing.
Secondly, since we increased the angle of attack it lead to an increase in the drag force faced by the car hence using the aspect ratio concept we were able to reduce the drag force faced by the rear wing.
Moreover we increased the angle of incline by 7 degrees of the front wing as from our CFD results noticed that in the earlier prototype a lot of the air was hitting the top of the wheel. Hence after the change were able to minimise the air hitting the wheel.

EVALUATION

The rear pods which were using weren't as effective as they were intended to be, not improving the airflow of the car by a significant margin. The centre of gravity of the car was not in an ideal position hence affecting the stability while racing. The curve present on the sidepod was not enough to direct the air to the area we intended it to go to, hence having a major effect on the flow of air as well.

10.4 /// FINAL **PROTOTYPE**

Drag Coeffiecient : 0.29
Lift Coefficient : 0.18
Mass of model : 44g



Development Goals from Prototype 2 :

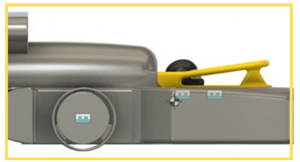
- Modification of rearpods back to the ones in Prototype 1 and make necessary modification
- Align the CoG to the thrust vector
- Modification to sidepods

/// Development of Goal 1

The rear pods' proposed shape would enhance airflow around the car's rear end, while also enabling the creation of a vortex behind the vehicle, which could drastically boost the speed. In prototype 1 the vortex was way to behind the car. We saw that by increasing the size of the point at the end of the vortex help us generate the vortex closer to the car also aiding us in generating downforce at the rear end. Evidence can be seen in section 4.4.

/// Development of Goal 2

From our research we saw we had to align the CoG horizontally with the thrust vector which was the CO2 chamber. In prototype 2 the CoG was coming a little lower than we wanted. Hence to keep it a little higher, we increased the diameter of the the body around the CO2 chamber by 1 mm. This meant there was going to be more mass at the top end hence the the CoG position went higher.



EVALUATION

We developed a wheel system which was efficient enough to reduce the rolling friction of the wheels, and also had a low moment of inertia. It also showed to have minimal imbalance.
The body we designed ensured good weight distribution. The sidepods reduced Magnus Effect induced lift. Vortexes were created, generating the required amount of downforce, thanks to the rear pods. Our front wing consisted of curved endplates, which were extremely beneficial to us as they deflected the air to the sides of the wheel, and the gradient of the wing redirected most of the air over the top of the wheel
The rear wing's NACA 0015 shape and incorporation of the correct aspect ratio significantly helped in reducing the drag of the car.